

1. Administrative & Study Identification

Full Study Title: Phase I Study of [177Lu]Lu-NNS309 in Patients With Pancreatic, Lung, Breast and Colorectal Cancers **Short Title/ Acronym:** [177Lu]Lu-NNS309 Phase I Solid Tumors **Protocol Number:** CFXX489A12101 **NCT Number:** NCT06562192 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [177Lu]Lu-NNS309 (Therapeutic) and [68Ga]Ga-NNS309 (Diagnostic/Imaging) **Phase of Development:** Phase I **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of [177Lu]Lu-NNS309 and to identify the recommended dose(s) (RD) and regimen(s) of [177Lu]Lu-NNS309 for further clinical evaluation. **Secondary Objectives:** To evaluate the preliminary anti-tumor activity of [177Lu]Lu-NNS309, characterize the pharmacokinetics of [177Lu]Lu-NNS309 in blood and urine, evaluate the radiation dosimetry for organs and tumor lesions, and evaluate the safety and imaging properties of [68Ga]Ga-NNS309.

2.2 Background & Rationale Fibroblast activation protein (FAP) is a type II integral membrane glycoprotein highly expressed on the cell surface of cancer-associated fibroblasts (CAFs) present in the tumor microenvironment of most epithelial cancers, while showing limited expression in normal tissues. Given this expression profile, FAP is a promising target for radioligand therapy with pan-cancer potential. [177Lu]Lu-NNS309 is a FAP-targeted radiopharmaceutical that shows improved tumor retention and demonstrates promising anti-tumor activity in preclinical models, addressing the need for targeted therapies in advanced refractory solid tumors.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation followed by dose expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Dose expansion will include ~94 patients (PDAC n~20, NSCLC n~20, HR+/HER2- ductal BC n~15, HR+/HER2- lobular BC n~15, and TNBC n~24) in addition to the dose escalation cohort. **Number of Sites:** Multi-center (Global across approximately 29 countries) **Estimated Study Start:** August

2024 **Estimated Primary Completion:** TBD (End of study involves a 36-month long-term follow-up period)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Locally advanced unresectable or metastatic pancreatic ductal adenocarcinoma (PDAC), non-small cell lung cancer (NSCLC), breast cancer (BC), or colorectal cancer (CRC) with disease progression following, or intolerance to prior standard-of-care therapies (e.g., cytotoxic chemotherapy, targeted therapy, or immunotherapy). 3. Patients must have all measurable lesions showing $[^{68}\text{Ga}]\text{Ga-NNS309}$ uptake on positron emission tomography/computed tomography (PET/CT).

4.2 Exclusion Criteria 1. Absolute neutrophil count (ANC) $< 1.5 \times 10^9/\text{L}$, hemoglobin $< 9 \text{ g/dL}$, or platelet count $< 100 \times 10^9/\text{L}$. 2. QT interval corrected by Fridericia's formula (QTcF) ≥ 470 msec or Creatinine clearance $< 60 \text{ mL/min}$. 3. Unmanageable urinary tract obstruction, urinary incontinence, or radiation therapy within 4 weeks prior to the first dose of $[^{177}\text{Lu}]\text{Lu-NNS309}$.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Escalating doses in Part 1 to determine the Recommended Dose (RD), followed by administration at the RD in Part 2 (Expansion). **Route of Administration:** Intravenous (IV) infusion for targeted radioligand therapy. **Duration of Treatment:** Administered in cycles until disease progression, unacceptable toxicity, or study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures for patients with advanced solid tumors. **Prohibited:** Concurrent systemic anti-cancer therapies, other investigational agents, or radiation therapy within 4 weeks prior to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Labs, $[^{68}\text{Ga}]\text{Ga-NNS309}$ PET/CT
Baseline	Day 0	Enrollment, First Dose of $[^{177}\text{Lu}]\text{Lu-NNS309}$	
Interim	Periodic Cycles	Safety Labs, PK & Dosimetry Assessments, Efficacy/Tumor Response Evaluation	End of Study
Up to 36 months	Long-term Follow-up for Disease Progression, Survival, and Radiation Safety		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability assessments to identify the recommended dose(s) (RD) and regimen(s), measured by the incidence of Dose-Limiting Toxicities (DLTs) and treatment-emergent Adverse Events. **Measurement Method:** Clinical evaluation and grading using CTCAE.

7.2 Secondary & Exploratory Endpoints - Preliminary anti-tumor activity (e.g., Objective Response Rate per RECIST 1.1). - Pharmacokinetics of [177Lu]Lu-NNS309 in blood and urine. - Radiation dosimetry for organs and tumor lesions.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite clinical visits, laboratory assessments, and continuous monitoring during dose escalation. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Safety Review Committee utilized specifically for evaluating DLTs and making cohort dose-escalation decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all patients receiving at least one dose; DLT-evaluable set for the dose-escalation phase. **Sample Size Justification:** Target n driven by standard Phase I dose-finding models followed by expansion cohorts ($\sim 15-24$ patients per tumor type) to estimate preliminary anti-tumor activity and further validate safety profiles. **Handling of Missing Data:** Follows standard Phase I oncology protocols (typically not imputed for primary safety endpoints).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of trial sites to ensure protocol adherence and patient safety. **Audit Trail:** eCRF locking, centralized review of PET/CT imaging, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CFXX489A12101 **Registry Number:** NCT06562192 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Phase I Study of [177Lu]Lu-NNS309 in Patients With Pancreatic, Lung, Breast and Colorectal Cancers **Official Regulatory Title:** A first-

in-human phase I, open-label, multi-center study of [177Lu]Lu-NNS309 in patients with pancreatic ductal adenocarcinoma (PDAC), non-small cell lung cancer (NSCLC), breast cancer (BC), and colorectal cancer (CRC).

12. Clinical Framework (Oncology/Radioligand Specific)

Primary Condition: Pancreatic, Lung, Breast, and Colorectal Cancers **Diagnosis Threshold:** Locally advanced unresectable or metastatic disease with documented [68Ga]Ga-NNS309 uptake on PET/CT **Medical Methodology:** FAP-Targeted Radionuclide Therapy (Radioligand Therapy) **Optimization Target:** Recommended dose (RD) and regimen identification for Phase II/III evaluation

13. Study Procedures & Methodology

Management Pathway: Radioligand Therapy Clinical Pathway **Education Protocol (HCP):** Radiopharmaceutical handling, dosimetry, and administration training **Education Protocol (Patient):** Radiation safety and post-administration precautions **Quality Audit Mechanism:** Central imaging review for [68Ga]Ga-NNS309 PET/CT screening and [177Lu]Lu-NNS309 dosimetry monitoring

14. Observation & Follow-Up Schedule

Total Duration: Up to 36 months long-term follow-up post-treatment **Onsite Visit Cadence:** Standard Phase I oncology visit schedule (Baseline, Day 1 of each cycle, End of Treatment) **Remote Monitoring Frequency:** As dictated by site-specific radioligand long-term follow-up protocols **Checklist Review Interval:** Every imaging/tumor assessment cycle (per RECIST 1.1)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				